

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

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**EC-7011-GS**

**B.E. VII Semester**

Examination, December 2020

**Grading System (GS)**

**Wireless Communication**

**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. Discuss about the spectrum limitations for wireless communication services.
2. Explain Knife edge diffraction model in detail.
3. A receiver is tuned to 1 GHz transmission and receives signals with Doppler frequencies ranging from 10Hz to 50 Hz when moving at a speed of 80 km/hr. What is the fading rate?
4. How does thermal noise calculate and discuss the noise figure and components to be considered in removal of receiver noise.
5. Discuss about the time domain measurements for channel sounding.
6. With suitable example explain Mathematical Model of Narrowband and wideband channels.

EC-7011-GS

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[2]

7. Determine the minimum bandwidth required to achieve a probability of error  $10^{-6}$  for an 8-PSK system operating at 20Mbps with carrier-to-noise ratio of 11dB.
8. Write short notes on any three of the following:
  - a) Frequency dispersive fading channels
  - b) Doppler spectra
  - c) Wave guiding
  - d) Micro and macro diversity
  - e) Narrow band model

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